Knockout Expression

The Genotype-to-Expression Strand: the Task, the Model, What the Campaign Established Round by Round, and What to Run Next

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Abstract

The task is to predict the $\log_2$ expression ratio of every measured gene in a single-deletion strain from the deletion alone, on the Kemmeren and Sameith compendia, scored by the per-gene Pearson across held-out strains against a replicate ceiling of 0.775. The incumbent, a graph-masked transformer whose encoder runs once on the wild-type graph and whose perturbation enters afterward through a one-key cross-attention, sits at 0.1965 ± 0.0222 across the eight long-budget arms of the v9 mask-schedule round, 25 % of the ceiling. Every score in this document is one statistic, the maximum of a centered 5-epoch rolling mean of the validation Pearson, read from the full history of every run and placed against those eight, whose spread bounds the replicate spread from above rather than measuring it. The campaign that followed the SIMB retrospective ran five rounds on IGB and Delta between 2026-08-31 and 2026-09-09.

What is established. The distributional head is not load-bearing: a point head collapses, the CRPS head collapses two times in three, and the Laplace-CRPS head ties the quantile head to 0.0005. Every decoder family tried before the campaign, the GEARS-style cross-gene readout among them, was stopped inside the first 300 epochs and was never compared. The validation loss bottoms out at a few hundred epochs in every run while the metric keeps rising to beyond 19,000, so the long budget buys Pearson on a rising objective. Training on the metric itself does not escape that: pure Pearson at batch 32 collapses to constant outputs at every seed, the batch-64 variant that survives peaks by epoch 2,400 on the long-budget band and gives part of it back, and a ranking objective (ListMLE) learns without collapsing but sits below the band at matched epochs so far. The one factor that moves the score by more than the replicate spread is the content of the gene embedding, ProtT5 over random by 0.07 at 1,000 epochs against a within-cell replicate sd of 0.025; trunk depth, readout width and weight decay do not. The per-gene readout is +0.03 over its in-round reference at two seeds, under the resolution of that round, and the pair term alone is a null. Four of four packed five-day tasks died of host memory.

What is not. Nothing measured beats the incumbent's mean at any budget; the incumbent's own embedding, `calm`, has not been placed against ProtT5; the CLS state is wild-type for every strain and no arm has yet put the perturbation inside the encoder; and no design run so far resolves a gap under about 0.04 at the full budget.

How to read this. section 1 is the strand as it stood before the campaign: the task, the model and where strain identity enters it, the heads and decoder families already tried, the published operators, and the imputation signal, carried over from the SIMB retrospective so nothing here needs that document. section 2 is the state of the strand claim by claim, with script, statistic and sample size. section 3 is the readout of the campaign's rounds. section 4 is what the evidence leaves open and what fits the hardware now free.

Working note: `notes/experiments.019-simb-multimodal.expression-strand-retrospective.md`

Generating scripts: `experiments/019-simb-multimodal/scripts/`

Leaderboard: `experiments/019-simb-multimodal/results/round_leaderboards.csv`

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

Contents

1 The strand before the campaign: task, model, and what was already known ✗

1.1 The task and its ceiling ✗

The **expression strand** predicts, for a strain carrying one gene deletion, the $\log_2$ ratio of every measured gene's transcript abundance against wild type. Targets are the Kemmeren 2014 knockout microarray compendium and the Sameith 2015 double-deletion compendium (`doi:10.1016/j.cell.2014.02.054` and `doi:10.1038/sdata.2015.15`), fused in the `fig3_core` build: 1,556 records, of which 1,484 are single deletions and 72 are Sameith doubles, so 95.4 % of the training signal carries exactly one perturbed gene. Every deleted gene appears exactly once, so under the strain-level partition no gene is in both the fit and the validation split and every validation strain is an unseen perturbation; a perturbation cannot be represented by its own observed response and must enter through an external gene embedding. SRC: `experiments/019-simb-multimodal/results/fig3_overlap_census.json`

The score is `pearson_per_feature`: for each reporter gene the Pearson correlation across held-out strains between predicted and measured $\log_2$ ratio, averaged over reporters. A model that predicts each gene's mean scores exactly zero on it, because a prediction constant across strains has no variance to correlate; the companion per-strain metric, `pearson_per_instance`, stays near 0.4 under that same collapse and is the form the perturbation-response literature reports (section 1.4). The **reproducibility ceiling** for the per-feature score is estimated from the 82 deletions measured independently in both compendia, as the mean over reporters of $\sqrt{\text{clip}(\hat{\rho}_g, 0, 1)}$ with $\hat{\rho}_g$ the reporter's test-retest correlation across the paired strains: **0.7746**. The agreement between the two studies' own measurements of those 82 deletions is 0.611, which is the ceiling for any cross-study statement. SRC: `experiments/019-simb-multimodal/results/expression_ceiling_replicate.json`

The byte count misleads. The compendium holds 9.2 million scalar labels against 0.39 million trigenic-interaction labels in the 010 queries, and the intuition that expression would therefore be the easier task was wrong. In the trigenic data a gene recurs across many records with different partners, so the supervision repeatedly constrains its contribution; in the expression data a gene is the perturbation exactly once and is measured on one array, so its 6,169 labels are 6,169 correlated readouts of a single perturbation event and nothing in the data isolates the deletion's effect from the array's. The perturbation axis has 1,484 points, and no objective or mechanism changes that.

1.2 The model, and the one place strain identity enters ✗

The incumbent is the equivariant cell-graph transformer (`equivariant_cell_graph_transformer.py` under `torchcell/models/`): gene tokens from a frozen embedding (`calm`) projected to width 90, a learnable CLS token prepended, six encoder layers whose self-attention is hard-masked by the nine interaction graphs, then a cross-attention **perturbation operator** whose keys are the deleted genes, then a per-gene readout MLP shared across every gene token. The checkpoint's state dict reconciles exactly to the logged parameter count.

module	parameters	role
`transformer_layers`	590,220	6 graph-masked encoder layers
`embedding_preprocessor`	369,639	input projection to $d = 90$
`post_perturbation_mixing`	101,430	Perceiver mixing after the perturbation
`perturbation_transform`	98,370	the cross-attention perturbation operator
`perturbation_head`	16,381	graph-level scalar head
`per_gene_head`	9,919	per-gene readout, $d \to 19$ quantile knots
`observed_label_encoder`	8,460	masked teacher-forcing encoder
`cls_token`	90	
total trainable	**1,194,509**	matches the logged `total_param_count`

Table 1: Parameter budget of the incumbent. The readout row is the striking one: 6,127 reporter genes are predicted by 9,919 parameters, because `PerGeneHead` is one MLP shared across every gene token. The perturbation operator, at 8 % of the parameters, carries the entire burden of making a prediction strain-specific.

The encoder runs once, on the wild-type graph, and every strain reuses it. The forward pass encodes the 6,607 gene tokens plus CLS once per batch, with no strain information in the input; the batch dimension appears only at the perturbation operator, which takes the shared gene states H and each strain's deleted-gene indices. Two consequences follow directly from the code. The CLS state h_{CLS} is wild-type for every strain: measured across-strain standard deviation 0.0 against 0.973 for the pooled perturbed-set summary z_S, so the conditioner input $[z_S; h_{\text{CLS}}]$ carries strain identity in one half only, and the FiLM arm was rewired to read z_S alone because half its first-layer weights saw a constant. And the perturbation never enters the graph-masked attention at all: the graphs shape how wild-type gene states mix, and the deletion is applied to the finished states afterward. SRC: torchcell/models/equivariant_cell_graph_transformer.py(PerGeneHeadconstructionandtheforwardpass;thesdfiguresarerecordedinthecodecommentthere)

At one deletion the operator has no pair term. The cross-attention's softmax runs over a key set of size one, so its weight is identically 1. Re-drawing W_Q and W_K at standard deviation 10 changes the output by exactly 0.0, and 16,200 of the operator's 32,760 attention parameters are dead at $|S_b| = 1$. The model therefore reduces to $g(h_i + c_b)$: reporter identity i and strain identity b meet once, by addition, and no term depends on the pair (deleted gene, reporter gene). Masked unmasking does not supply one where the score is taken: at reveal step $k = 0$ every mask bit is zero, the conditioning branch's inputs are zero as inputs, and the forward pass is identical to the unconditioned model. SRC: experiments/019-simb-multimodal/results/perturbation_selector_degeneracy.json

mechanism	pair rank	how one deletion reaches many genes
additive reference	0	it does not: one c_b added to every gene
null sink	9	sigmoid gate per head, α_i depends on the querying gene
graph propagation	data	random-walk mass from the deleted gene over the nine graphs, plus a hop-0 self indicator
low-rank bilinear	r	$\sum_j b_{ij}(h_i)\, a_j(c_b)$
response basis	r	factored readout, r shared response directions
GEARS-style cross-gene	r	pool over genes, low-rank broadcast back, in the decoder
per-gene readout	–	one output weight per reporter, as GEARS and State have
Hadamard / FiLM	90	$h_i \odot (1 + \gamma(c_b))$, gene and strain multiply

Table 2: Pair-term mechanisms resident in the model file, each gated to be the exact identity at initialization and selectable by config. The rank is what each can express, a derivation and not a measurement; which have been trained, and at what budget, is section 1.3 and section 3.2.

Rank is not what binds. The best rank-r approximation of the $1{,}482 \times 6{,}169$ target matrix, scored with the same per-feature Pearson on the matrix it was fit to, reaches 0.7265 at rank 32 and 0.7799 at rank 64, both at the ceiling. A rank-64 pair term has expressive room by a wide margin, so a null from such an arm is about the mechanism or the optimization and not about capacity. After removing each gene's mean, the residual gene-gene correlation pattern replicates on held-out strains at split-half $r = 0.8687$ against a permutation null of 8.5×10^{-5}, with effective rank 32.78; that structure is the reason a per-gene independent readout, which emits 6,169 separate marginals, discards most of what the data hold. SRC: experiments/019-simb-multimodal/results/lowrank_output_ceiling.json

SRC: experiments/019-simb-multimodal/results/residual_covariance_diagnostic.json

1.3 Heads and decoder families already trained ✗

Two comparisons are on record from before the campaign, and neither is a comparison.

Six distributional heads, five of them never past 200 epochs. The incumbent's quantile head rests on the table below, read from the leaderboard as it stood when the campaign was designed.

The decoder and perturbation families, all inside the early dip. The GEARS-style cross-gene readout, the rank-32 bilinear pair term, Perceiver post-perturbation mixing, the rank-32 response basis, context concatenation, graph propagation, the null sink and the deeper perturbation heads were each trained in waves 1 to 3 of project `torchcell_019_expr_v8`, most at one seed (table 4). Sixty-seven scored runs across waves 1 to 4b, and not one ran past 300 epochs, against a validation curve that dips at epochs 200 to 300 and then climbs for thousands more (section 2.5). Every one of them was stopped inside the dip, so the differences between arms are early-training transients and more seeds at that budget buy precision on the wrong estimand. Of the families in the table, only two have since been trained at a resolving budget: the rank-64 response basis and the per-gene readout, as the mechanism round (section 3.2). The GEARS-style cross-gene readout, the bilinear term, Perceiver mixing on its own and FiLM have not been.

Baselines that need no GPU, and one of them is a gate. Four were run before the campaign's first round and are the floor every arm is read against.

	all runs			matched budget, 100–200 epochs	
`dist` head	runs	best	max epochs	best	median
`quantile`	344	0.2382	9,900	0.1751	0.1352
`crps`	158	0.1631	199	0.1631	0.1267
`energy`	50	0.1421	199	0.1421	0.0960
`laplace_crps`	56	0.1411	199	0.1411	0.0775
`point`	34	0.1117	163	0.1117	0.1054
`nll`	59	0.0983	199	0.0845	0.0680

Table 3: The head comparison the incumbent rested on, before the objective round. The left block compares one head at 9,900 epochs against five at about 150; the right block is the window where all six ran, on which `quantile` leads by 0.0085 on the median with its maximum drawn from 84 runs against `crps`'s 25. `nll` is the designed σ-collapse control; `energy`'s covariance is global and cannot move a point metric. The objective round (section 2.2) is what replaced this table.

1.4 The published operators, and the benchmark that governs the reading ✗

Three papers were read in full against the campaign, mirrored and `sha256`-pinned. They converge on one point.

Every published operator in this family is additive. GEARS (Roohani et al. 2023, doi:10.1038/s41587-023-01905-6) combines a perturbation with a gene, and State (Adduri et al. 2025, doi:10.1101/2025.06.26.661135) forms its transformer input, as

$$h_u^{\text{post-pert}} = \mathrm{MLP}\big(h_u^{\text{gene}} + h^{\mathcal{P}}\big), \qquad \mathbf{H} = \mathbf{H}_{\text{cell}} + \mathbf{H}_{\text{pert}} + \mathbf{H}_{\text{batch}}.$$

CPA (Lotfollahi et al. 2023) is additive latent composition with gene specificity in the decoder. Neither Roohani paper ablates addition against a multiplicative or cross-attentional alternative, so the rank-0 operator of section 1.2 is the field's default and the structural finding is a statement about the model class.

Where they differ is the readout. Both put gene identity in per-gene output parameters applied to a shared nonlinear state: GEARS gives every gene its own linear layer $w_u \in \mathbb{R}^d, b_u \in \mathbb{R}$, and State's readout is one $\mathbf{W}_{\text{recon}} \in \mathbb{R}^{d_h \times G}$ with a column per gene. The incumbent composes to $F(h_i + c_b)$ with a single shared F, so reporter identity enters only through h_i, inside the same function the perturbation enters. A per-gene output weight costs $6{,}127 \times 91 = 557{,}557$ parameters here, +47 %, and it was promoted to the first mechanism arm on this evidence (section 3.2).

The benchmark. Ahlmann-Eltze, Huber and Anders 2025 (doi:10.1038/s41592-025-02772-6) benchmarked five foundation models and two purpose-built ones, GEARS and CPA, against deliberately simple baselines: "None outperformed the baselines." On unseen single perturbations, "None of the deep learning models was able to consistently outperform the mean prediction or the linear model." Their linear model, $\mathrm{argmin}_{\mathbf{W}} \|\mathbf{Y}_{\text{train}} - (\mathbf{GWP}^{\top} + \boldsymbol{b})\|_2^2$, is `B2` above, and it beat GEARS while using GEARS's own perturbation embeddings. State reports the same of itself on genetic perturbations: "On genetic perturbation datasets, State matched the performance of the perturbation mean baseline", a 600M-parameter encoder trained on 167 million cells at parity with a mean. Two consequences: scale is not the lever, which the campaign's own record agrees with at 145 runs from 0.12M to 4.7M parameters and `pearson(log10params,score)` $= -0.129$; and the "no better than the mean" squared-error result is the field's central finding stated honestly rather than a local defect. The comparability caveat cuts the other way: their Pearson is computed across genes within one perturbation, where a mean baseline scores high, so the incumbent's 0.196 and their 0.556 are different statistics and any comparison must use `pearson_per_instance`.

What is not portable. State's contribution is its set machinery, bidirectional attention across sets of 256 cells trained with an energy-kernel MMD, which needs a population of measurements per condition. With one bulk profile per deletion strain, at $S = 1$ that MMD is $2\|\hat{x} - x\|_2$, an ordinary per-sample loss, and cross-cell attention has nothing to attend to. The ceiling rests on 82 cross-study shared deletions, not on per-strain replicates, which is also the argument for replicate measurement in any screen this group designs.

1.5 Imputation is the largest signal in the data, and it is orthogonal ✗

A Gaussian conditional mean with a ridge loading, fit once on training strains and never trained by gradient, that observes m measured genes of a held-out strain and predicts the rest reaches far past anything the model does. At $m = 0$ it predicts the per-gene mean and scores exactly 0, which is why it is a measurement of information and not a rival model: the model is scored at $m = 0$, where the oracle has nothing. SRC: `experiments/019-simb-multimodal/scripts/masked_conditioning_oracle.py`

Knowing which gene was deleted and knowing some of this strain's other measured genes are almost separate sources of information about the rest, and the strand is scored where only the first is available. The parameter-free companion on

Table 4: The decoder and perturbation-family arms of waves 1 to 3 in project `torchcell_019_expr_v8`, 35 runs over 21 arm-wave cells, scored by the maximum of a centered 5-epoch rolling mean of the validation Pearson (`score_decoder_arms.py`). Every run stopped between epochs 50 and 276, inside the early dip of a curve that is still rising at 9,900 (section 2.5), so the scores measure early-training transients and the column is not an arm comparison. The GEARS-style cross-gene readout, the bilinear pair term, Perceiver mixing and the response basis are all here, each at one seed. The mechanism column is a gloss on the override, which is the launcher's own text.

Arm	Wave	Mechanism (override)	Seeds	Epochs	Smoothed max, mean / best
`A0_baseline`	1	additive reference, $g(h_i + c_b)$ (`none`)	3	80–120	0.081 / 0.153
`A1_free16`	1	free per-gene lookup of 16 dims (`multitask.free_gene_dim=16`)	1	139	0.152 / 0.152
`A2_self`	1	self-indicator only (hop 0) (`"$PROP=true" model.perturbation_propagation.hops=0`)	1	123	0.152 / 0.152
`A4_prop_h2`	1	random-walk propagation, 2 hops (`"$PROP=true" model.perturbation_propagation.hops=2`)	3	70–140	0.100 / 0.172
`A6_prop_sparse`	1	propagation, 2 hops, sparse (`"$PROP=true" model.perturbation_propagation.hops=2 "model.perturbation_propagation.graphs=[regulatory_interaction,tflink,physical_interaction]"`)	1	140	0.175 / 0.175
`B0_protT5`	1	ProtT5 gene embeddings (`"cell_dataset.node_embeddings=[prot_T5_all]"`)	1	70	0.089 / 0.089
`A6_prop_sparse`	2	propagation, 2 hops, sparse (`"$PROP=true" model.perturbation_propagation.hops=2 "model.perturbation_propagation.graphs=[regulatory_interaction,tflink,physical_interaction]"`)	2	76–88	0.068 / 0.090
`C0_concat`	2	head sees $[h_{\text{pert}}; h_i; c_b]$ (`multitask.concat_context=true`)	1	140	0.161 / 0.161
`D1_bilinear32`	2	rank-32 bilinear pair term (`multitask.bilinear_rank=32`)	1	123	0.152 / 0.152
`E0_perceiver32`	2	Perceiver mixing after the perturbation (`model.post_perturbation_mixing.enabled=true model.post_perturbation_mixing.num_latents=32`)	1	114	0.157 / 0.157
`A0_baseline`	3	additive reference, $g(h_i + c_b)$ (`none`)	1	196	0.158 / 0.158
`A1_ref`	3	additive reference, $g(h_i + c_b)$ (`none`)	4	242–250	0.076 / 0.158
`A2_prop_sparse`	3	propagation, 2 hops, sparse (`"$PROP=true" model.perturbation_propagation.hops=2 "model.perturbation_propagation.graphs=[regulatory_interaction,tflink,physical_interaction]"`)	4	50–51	0.079 / 0.130
`A4_nullsink`	3	null-sink gated attention (`model.perturbation_head.null_sink=true model.perturbation_head.null_sink_bias_init=-4.0 model.perturbation_head.null_sink_trainable=true`)	4	186–189	0.076 / 0.149
`D1_bilinear32`	3	rank-32 bilinear pair term (`multitask.bilinear_rank=32`)	1	175	0.148 / 0.148
`D2_mask`	3	hard graph mask replaces the KL (`model.attention_mask.enabled=true model.graph_regularization.graph_reg_lambda=0.0`)	1	235	0.149 / 0.149
`E0_perceiver_on`	3	Perceiver mixing after the perturbation (`model.post_perturbation_mixing.enabled=true model.post_perturbation_mixing.num_latents=32 'model.post_perturbation_mixing.gate_mode=on'`)	1	181	0.127 / 0.127
`F1_attn4`	3	deeper perturbation head (`model.perturbation_head.num_layers=4 model.perturbation_head.extra_layer_ffn_mult=0`)	1	222	0.155 / 0.155
`F1_deep2`	3	deeper perturbation head (`model.perturbation_head.num_layers=2`)	1	202	0.168 / 0.168
`GEARS_crossgene`	3	GEARS-style cross-gene readout, rank 64 (`model.cross_gene.enabled=true model.cross_gene.rank=64`)	1	168	0.133 / 0.133
`H0_factor`	3	rank-32 response basis (factored readout) (`multitask.response_basis_rank=32`)	1	276	0.155 / 0.155

	prediction	what it isolates	pf	nmse
B0	each gene's training mean	the definitional floor	0.0000	1.0000
B1	no change, $\log_2$ ratio $= 0$	"nothing happened"	0.0000	1.3784
B2	$\mathbf{GWP}^\top + \boldsymbol{b}$, ridge, rank-swept	a pair term without deep learning	**0.1040**	–
B3	mean of the k nearest deleted genes in embedding space	genotype-side signal from proximity alone	0.0908	–

Table 5: The four baselines, measured 2026-08-31, `pf` = `pearson_per_feature`. `B2` and `B3` are the best over four gene embeddings, both won by `prot_T5_all`. `B0` and `B1` are definitionally zero and `B0`'s `nmse` of exactly 1.0000 is the quantity `nmse` divides by, so the two are the data-path self-check. The incumbent's 0.1965 ± 0.0222 clears `B2` by 0.093, about 4.2 standard deviations, which is the opposite of what the published benchmark found for GEARS and the reason the campaign's head question was worth its GPUs. SRC: `experiments/019-simb-multimodal/scripts/expression_baselines.py`

	tokens	perturbation enters	per-gene readout
GEARS	genes	$\mathrm{MLP}(h_u + h^{\mathcal{P}})$	yes, $w_u \in \mathbb{R}^d$
State	cells	$\mathbf{H}_{\text{cell}} + \mathbf{H}_{\text{pert}}$	yes, $\mathbf{W}_{\text{recon}} \in \mathbb{R}^{d_h \times G}$
torchcell v9	genes	$g(h_i + c_b)$	no, one shared MLP

Table 6: Where gene specificity lives in the three operators. These are literature descriptions, not measurements made here.

the genotype side, a similarity-weighted average of the nearest other deleted genes' phenotypes in `prot_T5_all` space, reaches 0.117 against a random-embedding floor of 0.033, so over half the genotype-side signal the trained model finds is available from embedding proximity alone. SRC: `experiments/019-simb-multimodal/results/knn_embedding_probe.json`

1.6 The campaign's arithmetic ✗

One run is a multi-day object. The v9 `M_fine` run took 91.4 hours for 9,999 epochs at 32.9 s/epoch and peaked at epoch 9,674; the two IGB card classes deliver the same wall clock, 31.0 and 30.5 s/epoch on RTX 6000 Ada and A100, because validation and checkpointing dominate compute, so 9,900 epochs is 3.5 days solo on either and the partition can be chosen on availability. Packing two runs on one card was measured, not assumed: 1.53× per-run slowdown for 1.31× aggregate throughput at 9.5 GiB of VRAM per run, so a packed run reaches about 9,100 epochs in five days. What packing did not measure was host memory, and section 3.5 is the cost of that. SRC: `experiments/019-simb-multimodal/results/expression_objective_diagnosis.json`

SRC: `experiments/019-simb-multimodal/results/launch_plan_evidence.json`

The launch was sized at $\sigma = 0.0222$, the spread of the eight long-budget arms, which section 2.1 shows is an upper bound on the replicate spread. At that figure a symmetric two-arm contrast needs 19 runs a side to see 0.02 and 5 to see 0.04, while one fresh arm of three against the eight-run baseline sees 0.042; the matched-budget head gap on record was 0.0085, so no affordable design resolved the real gap and the objective round was built to detect a large effect or report indistinguishability. The replicate spreads measured since, 0.012 to 0.025 within cell at 1,000 epochs and 0.019 at 4,000, make every design in section 4 cheaper than that arithmetic said.

m revealed	within-study	Kem → Sam	% of its m=1000	gain surviving a genotype predictor
10	0.4201	0.2229	46 %	97.5 %
100	0.6869	0.3668	77 %	99.2 %
1000	0.7977	0.4793	100 %	100.6 %

Table 7: The masked-conditioning oracle, `pearson_per_feature` on validation strains, genes revealed at random. The cross-study column is the honest one, since within-study conditioning partly predicts the target array's own noise; against the 0.611 cross-study ceiling, 100 genes recover 60 %. The last column replaces the per-gene mean with a k-nearest-neighbor genotype predictor over `prot_T5_all` and re-runs the oracle on its residuals: the gain does not shrink, so the two channels are nearly non-overlapping. Choosing the revealed genes by across-strain residual variance beats random at the cheap end by +0.046 at $m = 10$ across studies.

2 The state of the strand, claim by claim ✗

The task is the one of section 1.1, scored by the per-gene Pearson across held-out strains and averaged over genes against a replicate ceiling of 0.775. Everything below is read from the leaderboard puller (`pull_round_leaderboards.py`) over project `torchcell_019_expr_v9`, which is the masked-label objective, or from a script named in the paragraph. The statistic throughout is `roll_max`: the maximum of a centered 5-epoch rolling mean of `val/expression/pearson_per_feature`, an upward-biased order statistic whose bias grows with the epochs run, and which cancels in a contrast between runs scored the same way.

2.1 The incumbent and its spread ✗

The incumbent is the quantile head under the pinball loss, learning rate 3×10^{-4}, dropout 0.1, six transformer layers at hidden width 90, the nine-graph attention mask, `calm` gene embeddings, batch 32, 9,900 epochs, and the reveal schedule $[0, 10, 100, 1000]$. Eight runs at that budget share every one of those settings the leaderboard records and score $\mathbf{0.1965 \pm 0.0222}$; they were read as identical-config replicates until 2026-09-08, when their W&B configs showed them to be the eight arms of the v9 mask-schedule round (`M_sched`, the incumbent schedule; `M_lo`, `M_hi`, `M_fine`, `M_coarse`, other schedules; `M_nomix`, mixing off; `M_gate_rezero`, gate closed; `M_off`, no schedule). So 0.1965 ± 0.0222 is a mean and spread across arms, the spread bounding nondeterminism from above; the incumbent schedule itself is one draw at 0.1824, and the three highest leaderboard entries, 0.238 to 0.243, are the `M_fine` arm and its resumed lineage to 19,000 epochs, one draw of one schedule. Replicate spread measured on runs that do share a configuration is 0.0246 pooled within cell at epochs $\leq$ 990 (section 3.1) and 0.019 between two seeds of the mechanism-round reference at epochs $\leq$ 4,079 (section 3.2). Baselines that need no GPU (`expression_baselines.py`): per-gene mean 0.0000, bilinear on ProtT5 0.1040, embedding-neighbor 0.0908. The incumbent clears the bilinear by 0.093, about 4.2 standard deviations. On squared error no live run beats the per-gene mean at its Pearson peak: the best `nmse` at peak is 1.005.

2.2 The head is not load-bearing ✗

The objective round (IGB, launched 2026-08-31, jobs 2368333, 2368337, 2368339; three replicates per challenger at the incumbent's budget) asked whether the quantile head does anything a point head or another proper scoring head would not.

head	n	collapsed	live mean	live sd	live max	min `nmse` at peak	epochs
`point`	6	6	–	–	–	–	2,039–9,895
`crps`	6	4	0.1697	0.0094	0.1764	1.075	1,995–9,896
`laplace_crps`	6	0	0.1947	0.0167	0.2154	1.025	1,995–9,896
`quantile`	20	0	0.1960	0.0253	0.2430	1.005	1,593–18,990
`pearson`	5	3	0.1851	0.0067	0.1899	1.365	3,162–9,107, running
`pearson_mse`	3	2	0.1859	–	0.1859	1.046	3,162–6,573, running
`listmle`	4	0	0.1508	0.0116	0.1676	74.7	1,789–2,430, running

Table 8: The objective round (upper block) and the two metric-aligned rounds that followed it (lower block), on the leaderboard's rule, which reads a 500-sample history and so places each score slightly below the full-history readouts of section 3. Collapse is a final Pearson numerically zero: the network emits one constant per gene. The point head collapses every time, CRPS two times in three, the Laplace-CRPS and quantile heads never. Between the two that survive the gap is 0.0005 against a detectable gap of about 0.042 at three replicates versus the eight long-budget arms at their spread of 0.0222, an upper bound on the replicate spread. The lower block is partial: every one of those jobs was still running on IGB at the read, and the two `listmle_mse` runs that collapsed and were killed are below the 1,000-epoch filter. `listmle`'s `nmse` of 74.7 is not a defect of the fit: the ranking objective is shift-invariant, so each gene's predicted location drifts freely while its ordering, which the metric reads, does not.

What the head buys is not collapsing. Between the two heads that do not collapse the choice was never load-bearing, and the round was sized so that only a large effect could have shown; the honest statement is that the heads are indistinguishable at any budget that can be afforded. This is the head question; the decoder-family question, whether a GEARS-style cross-gene readout, a bilinear pair term or Perceiver mixing changes the score, has never been asked at a resolving budget (section 1.3, table 4), and the one family that has, the per-gene readout, is section 3.2.

What the incumbent's head optimizes and what the metric reads are different things (fig. 1). The head emits 19 quantile knots per gene on the grid $\tau = 0.05, 0.10, \ldots, 0.95$; each knot has its own pinball term $\rho_\tau(u) = \max(\tau u, (\tau - 1)u)$ on its own residual $u = y - \hat{q}_\tau$, and the objective is their mean. `pearson_per_feature` reads the median knot alone. On one illustrative gene the median term is 8.1 % of the loss, so the other 18 knots carry 91.9 % of the gradient through the shared trunk toward a quantity the metric never inspects. The knots are 19 points on the predictive CDF, so they also carry a per-gene noise estimate and a calibration check: the fraction of observations that fall inside the central 50 % and 80 % intervals. At epoch 10,000 the incumbent reads 0.326 and 0.578 against 0.50 and 0.80, both far below the diagonal, and both back-solve under a Gaussian scale mismatch to the same predicted-over-true width ratio, 0.624 and 0.627: the intervals are 1.6× too narrow. Simulating that one ratio reproduces the two measured coverages. The spread that most of the loss is spent on is itself miscalibrated, in the overconfident direction, while the point estimates are over-dispersed at $s/r = 1.92$ (SIMB retrospective, campaign section).

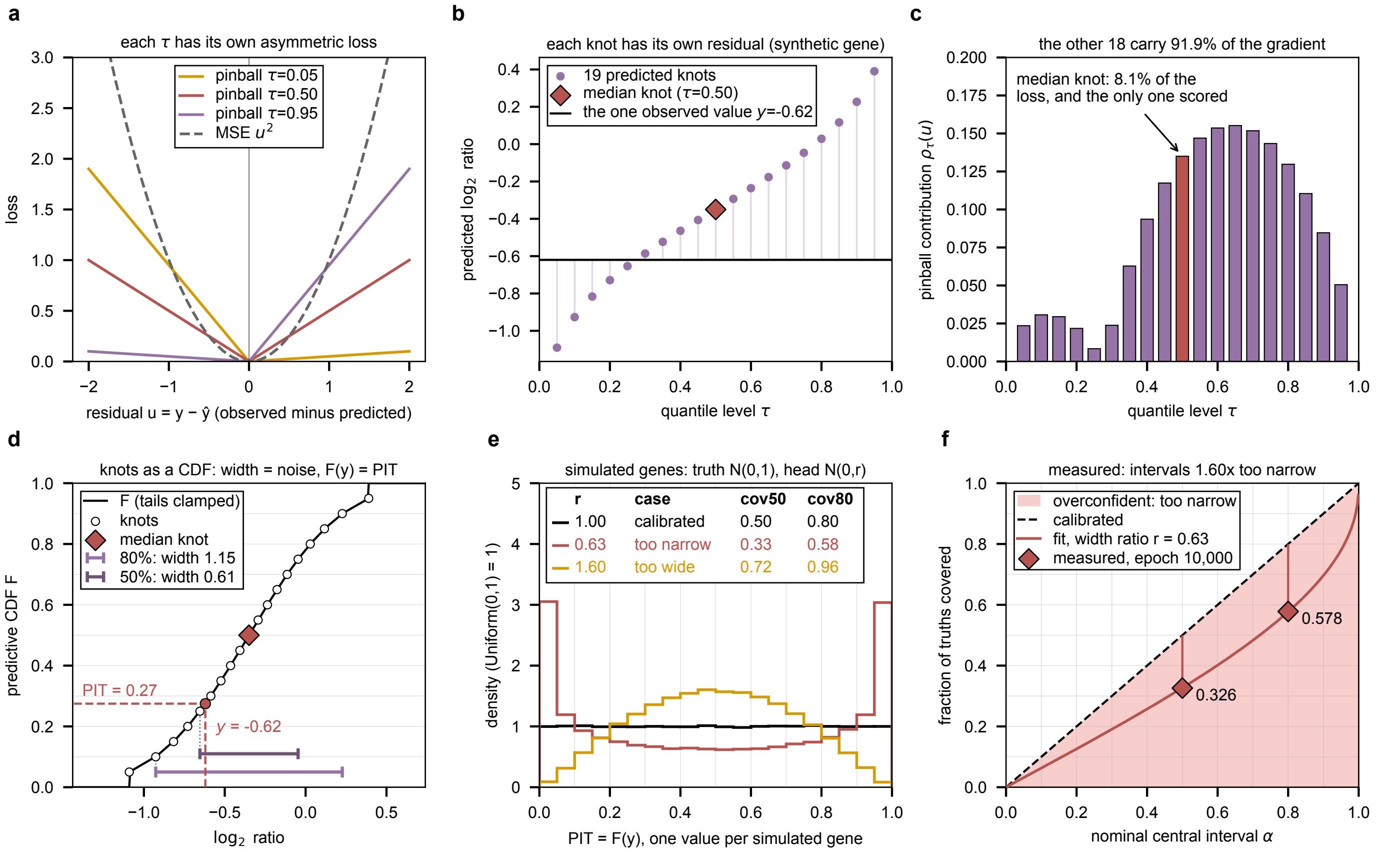

Figure 1: The pinball loss, what its knots carry, and what the metric reads. Panels **a** to **e** are synthetic; **f** is measured. **a**, the pinball loss at three quantile levels against squared error. **b**, one illustrative gene: 19 predicted knots, the median knot, and the one observed value; each knot's residual to that value feeds its own pinball term. **c**, the 19 terms; the median carries 8.1 % and is the only knot the metric scores. **d**, the same knots read back as a CDF (joined piecewise-linearly, tails clamped as `_quantile_pit` does): the central 80 % and 50 % interval widths are the model's own noise estimate for that gene, and the CDF at the observed value is its PIT. **e**, PIT histograms over 200,000 simulated genes with truth $N(0,1)$ and a head predicting $N(0,r)$: flat when $r = 1$, U-shaped when the intervals are too narrow, humped when too wide, with the coverage of the central 50 % and 80 % intervals per case. **f**, the incumbent's measured coverage at epoch 10,000 as a reliability curve; the fitted curve is a single width ratio $r = 0.63$, derived not measured. Generated by `plot_pinball_vs_mse.py`.

2.3 Spread across the long-budget arms is a function of budget, and not a monotone one ✗

The same eight long-budget arms, rescored on the prefix of their own curves (`short_budget_spread.py`). Read every spread below as an arm spread plus nondeterminism, an upper bound on the replicate spread at that budget:

Two traps were hit measuring this and are recorded so the next reader does not hit them: a resumed run passes an epoch filter and inherits its parent's config, so it masquerades as a replicate and is identified only by where its curve starts; and a whole-curve-maximum collapse test reads a run that collapses late as healthy, so collapse is tested on the last value.

2.4 A short screen does not rank like a long run, within one configuration ✗

Spearman between the 700-epoch and 3,000-epoch `roll_max` of the same run (`budget_rank_preservation.py`): −0.139, $p = 0.53$, $n = 23$; top-3 overlap one of three, top-5 one of five. After resume exclusion the corpus is the eight mask-schedule arms plus the objective round, so this measures persistence among near-identical configurations: a run's early

epochs	mean	sd	main effect, 32 v 32	cell, 16 v 16	arm, 8 v 8
350	0.1198	0.0096	0.0067	0.0095	0.0135
500	0.1223	0.0058			
700	0.1411	0.0166			
1,000	0.1609	0.0099			
1,500	0.1670	0.0117			
2,000	0.1745	0.0151			
2,800	0.1821	0.0175			
4,000	0.1883	0.0171			
6,000	0.1917	0.0177			
9,900	0.1965	0.0222			

Table 9: Mean and spread of the eight long-budget v9 arms at truncated budgets, from 500-sample curves. The spread is not monotone in budget: 700 epochs is the noisiest point below 2,000. The last three columns are the detectable gap at the 350-epoch spread for the three contrast sizes a 2^4 grid affords, the same arithmetic that sized the v10 grid at 1,400 epochs and two seeds rather than 700 and four.

rank among them says nothing about its late rank. Whether a short screen orders different configurations correctly is answerable only on a multi-config corpus, and the v10 grid (section 3.1) is the first one.

2.5 The validation loss bottoms out early and the metric keeps rising ✗

Full per-epoch history of every live v9 run whose loss is not the metric, 27 runs of 1,000 epochs or more that neither collapsed nor resumed, read 2026-09-09 (`loss_min_vs_pearson_peak.py`, fig. 2); the four `listmle` runs that joined the objective round after table 8 was read are included, and both `listmle_mse` runs collapsed. The median epoch of the `val/loss` minimum is 481 (range 203 to 3,097) and no run bottoms out by epoch 100. The median final loss sits 9.0 % above its minimum. Every run's Pearson peak (median epoch 2,433) comes after its loss minimum, at a loss already 6.8 % above the floor, and the median Pearson gained between the two is +0.057. The long budget buys Pearson on a rising validation objective, which is the fact the metric-aligned round was built to confront. The eight Pearson-objective runs are excluded here by construction, since their loss and metric coincide on train.

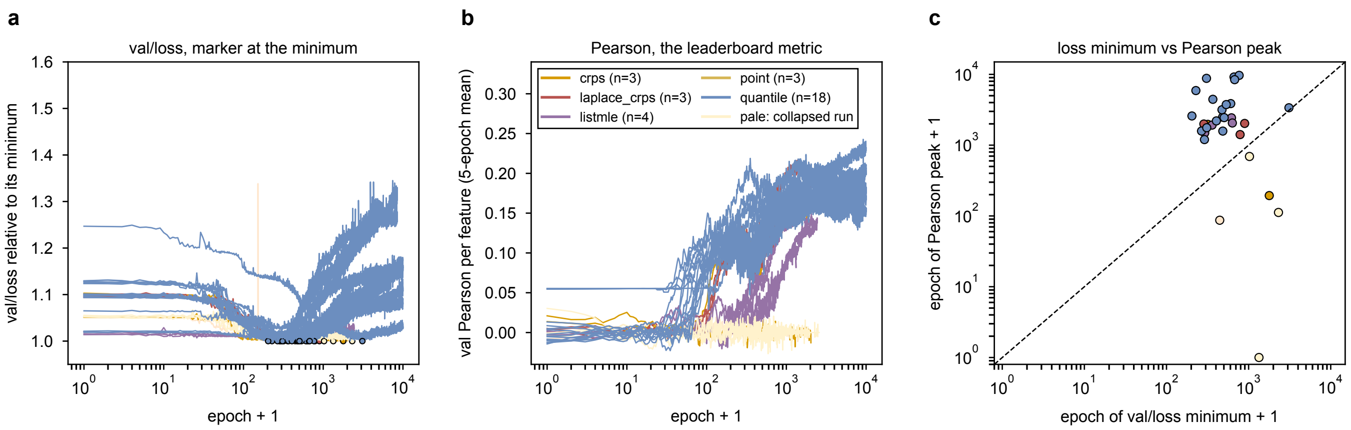

Figure 2: Where the loss bottoms out and where the Pearson peaks. **a**, validation loss relative to its minimum, one curve per run, colored by head; pale curves are collapsed runs. **b**, the 5-epoch rolling mean of validation Pearson. **c**, the epoch of the loss minimum against the epoch of the Pearson peak; every point lies above the diagonal. Generated by `loss_min_vs_pearson_peak.py`.

2.6 Every run behind these numbers ✗

Table 10 links every run this document draws a number from to its W&B page, generated by `wandb_run_index.py` from the same result files the sections read.

Table 10: Every run behind the numbers in this document, 80 in all, in projects torchcell_019_expr_v9 and torchcell_019_expr_v10 (entity zhao-group). One row per round and arm; each entry is a seed and its run id, linked to the run page, with the final epoch in brackets. The objective round lists each fresh run and its resume. The v10 grid names cells by embedding, trunk, readout and weight decay.

Round	Arm	Seed: run [epochs]
v9 mask arms	M_coarse	0: ebkzn1ao [9,900]
	M_fine	0: hx8pxdic [9,900]
	M_gate_rezero	0: u1vuznme [9,900]
	M_hi	0: da5g4o9v [9,900]
	M_lo	0: f2wf23oy [9,900]
	M_nomix	0: rb3bhryq [9,900]
	M_off	0: tow1z48n [9,900]
	M_sched	0: 8r5ewoaq [9,900]
objective	Q_point	0: 8gklzy7n [9,895] 0: iiqohck6 [2,039] 1: yrs7x07y [9,895] 1: qplil73c [2,039] 2: y8ex71vn [9,863] 2: he2vb4qr [2,577]
	Q_crps	0: m6utaeu8 [9,896] 0: lljkhhck [1,995] 1: djgkdzol [9,895] 1: nx5dwhp8 [2,039] 2: eiu664kx [9,888] 2: hzumvs8y [2,031]
	Q_laplace	0: 1xa3x8uc [9,896] 0: c981k5n1 [1,995] 1: 7v9l6ett [9,895] 1: ia6312dv [2,039] 2: g69wskm0 [9,888] 2: lz8gq5ui [2,031]
mechanism	R_ref	0: i63dbvha [4,139] 1: r2s57bvt [8,499]
	R_basis64	0: h9l8aguy [8,499] 1: 4gtnrja7 [7,018]
	R_pergene	0: 4u0txt3s [8,499] 1: vqek7ali [8,499]
	R_pergene_basis64	0: nrkorko0 [4,079] 1: 3qy1rh0o [6,589]
metric-aligned	Q_pearson	0: zaul43n1 [4,298] 1: kspeoljg [4,291] 2: p2qq2525 [3,166]
	Q_pearson_mse	0: hsauh7al [4,298] 1: ppc2pyv5 [6,576] 2: 4ei42tb1 [3,169]
	Q_pearson_b64	0: 7ylecrjz [9,118] 1: wb2xocf2 [9,114]
ranking	Q_listmle	0: teaeym0k [1,959] 1: g755dzhn [1,792]
	Q_listmle_mse	0: dhqpktod [637] 1: j6m31tov [480]
	Q_listmle_b64	0: k79hvcqe [2,438] 1: 6b33ftkt [2,437]
v10 grid	c0_rand_big_mlp_wd0	0: dd26f5nw [1,399] 1: nxrpgwca [1,176]
	c1_ptt5_big_mlp_wd0	0: te8272kk [1,399] 1: l4uw0g1e [1,399]
	c2_rand_small_mlp_wd0	0: j5oc0kbf [1,399] 1: zu0mp0ly [1,399]
	c3_ptt5_small_mlp_wd0	0: lnoei9xt [1,399] 1: wv0qlp4n [1,192]
	c4_rand_big_lin_wd0	0: 0htrkihg [1,119] 1: 59a4xytq [1,399]
	c5_ptt5_big_lin_wd0	0: 6ee07w1j [1,399] 1: 7ue6aduu [990]
	c6_rand_small_lin_wd0	0: 6ktjsou6 [1,399] 1: ujbunglx [1,020]
	c7_ptt5_small_lin_wd0	0: dcx8l9z6 [1,148] 1: ts735yhp [1,399]
	c8_rand_big_mlp_wd2	0: fk6luj3g [1,119] 1: 7n0ohmht [1,399]
	c9_ptt5_big_mlp_wd2	0: 5ivl9nat [1,399] 1: iyvyd7c3 [990]
	c10_rand_small_mlp_wd2	0: p2dj91nf [1,399] 1: yecoyfgc [1,029]
	c11_ptt5_small_mlp_wd2	0: oe77kyf6 [1,148] 1: hhvof18w [1,399]
	c12_rand_big_lin_wd2	0: kt9tezrk [1,399] 1: rd64e6kx [1,170]
	c13_ptt5_big_lin_wd2	0: r9w7i1fw [1,399] 1: c3xrtsr7 [1,399]
	c14_rand_small_lin_wd2	0: jrwpcoxd [1,399] 1: 4ity6oib [1,399]
	c15_ptt5_small_lin_wd2	0: a6mmvj8i [1,399] 1: 04204r3z [1,198]

2.7 Training on the metric does not resolve it ✗

Two heads were added to `torchcell/losses/distributional.py` on 2026-09-06: `pearson`, whose loss is one minus the mean per-feature Pearson over the genes hidden at the current unmasking step, each gene's correlation taken over the strains in the batch where it is hidden, with columns of fewer than three scored rows or constant prediction or target dropped as the metric drops them; and `pearson_mse`, the same plus the masked squared error at unit weight as a scale anchor. Both are point-shaped, carry no PIT and no calibration keys, and the pure loss is scale-free, so its `mse`, `nmse` and spread ratio are not comparable to any other arm's. Eighty-six unit tests and a container canary (`2375312`) preceded the launch. The readout is section 3.3: collapse at every batch-32 seed of the pure loss, at two of three seeds of the anchored one, and parity with the incumbent for the batch-64 survivors, which peak by epoch 2,400 and give part of it back.

A third pair followed on 2026-09-08: `listmle`, the Plackett-Luce negative log-likelihood of the true ordering of the strains in the batch, per hidden gene, divided by the list length (Xia et al. 2008), and `listmle_mse`, the same plus the masked squared error at unit weight. The question it asks is the ranking one: given a set of knockouts, which strain expresses gene g highest. ListMLE keeps a gradient on a constant column ($\log(n!)/n$ per gene, where Pearson drops the column), is shift-invariant but not scale-invariant, so predicted spread can only grow under it, and makes the batch part of the objective as Pearson does. Ninety-eight unit tests and a canary (`2385788`) preceded the launch; the readout so far is section 3.4.

3 The readouts: four rounds, one large effect ✗

Three rounds returned within two days of each other on 2026-09-08, the 2^4 factorial on Delta, the mechanism round on IGB (both sized by the arithmetic of section 1.6) and the metric-aligned objective round, which trains on $1-r$ directly (section 2.7); the ranking round that replaced the dead half of the metric-aligned one is a day old at this read. Every score below is `roll_max`, the maximum of a centered 5-epoch rolling mean of `val/expression/pearson_per_feature`, read from the full per-epoch history, and every contrast is at a MATCHED budget computed as the smallest final epoch across the runs being compared. The scoring rule is an upward-biased order statistic; it is the same rule on every side of every contrast, so the bias cancels in a difference and not in an absolute number. Runs still training are read at the epoch they had reached, stated with each number, and W&B's `finished` state is not evidence that a run ended: these runs train offline on IGB and every sync from the login node stamps the snapshot `finished`. Whether a job is alive is read from `squeue`.

3.1 The factorial: embedding content is the only factor that moves the score ✗

Job `21830323` ran 16 cells at two seeds, one run per GPU, four per node. Five of the eight array tasks completed 1,400 epochs; three hit the two-day wall between epochs 990 and 1,198, so the matched budget is epochs ≤ 990.

factor	contrast (level 1 − level 0)	effect	t	effect, 31 runs	t
embedding	`prot_T5_all` − `random_1024`	+0.068	+7.8	+0.076	+17.3
trunk	$L{=}2, h{=}45$ − $L{=}6, h{=}90$	−0.012	−1.4	−0.018	−4.0
readout (inert, see text)	linear − two-layer	+0.007	+0.8	+0.002	+0.5
weight decay	10^{-4} − 10^{-8}	+0.005	+0.5	−0.000	−0.1
embedding × trunk		+0.017	+2.0	+0.013	+2.9

Table 11: Main effects of the v10 grid at epochs ≤ 990, 16 runs a side. The error is the pooled within-cell replicate standard deviation, 0.0246 on 16 degrees of freedom over all 32 runs (standard error per effect 0.0087), and 0.0122 on 15 degrees of freedom once the one run that never left the chance band is dropped (standard error 0.0044). The other five two-way interactions have $|t| < 1.3$ in both readings.

Embedding content is worth 0.07, three times the replicate spread. The ProtT5 cells average 0.129 against 0.061 for random vectors of the same width, and no cell of one level overlaps any cell of the other (fig. 3a). The contrast was built as a content test at matched width 1,024, so it says the content matters and says nothing about `calm`, the incumbent's embedding, which is not a level of the grid. The one healthy draw of the incumbent-with-ProtT5 cell scores 0.141 against the incumbent's 0.161 ± 0.010 at the same budget, one draw against eight, and that is not a resolved gap in either direction.

Weight decay is a measured null; the readout row is retracted. The weight-decay effect is under 0.008 against a resolution of about 0.02 at two standard errors. The readout row is a null by construction, not a measurement:

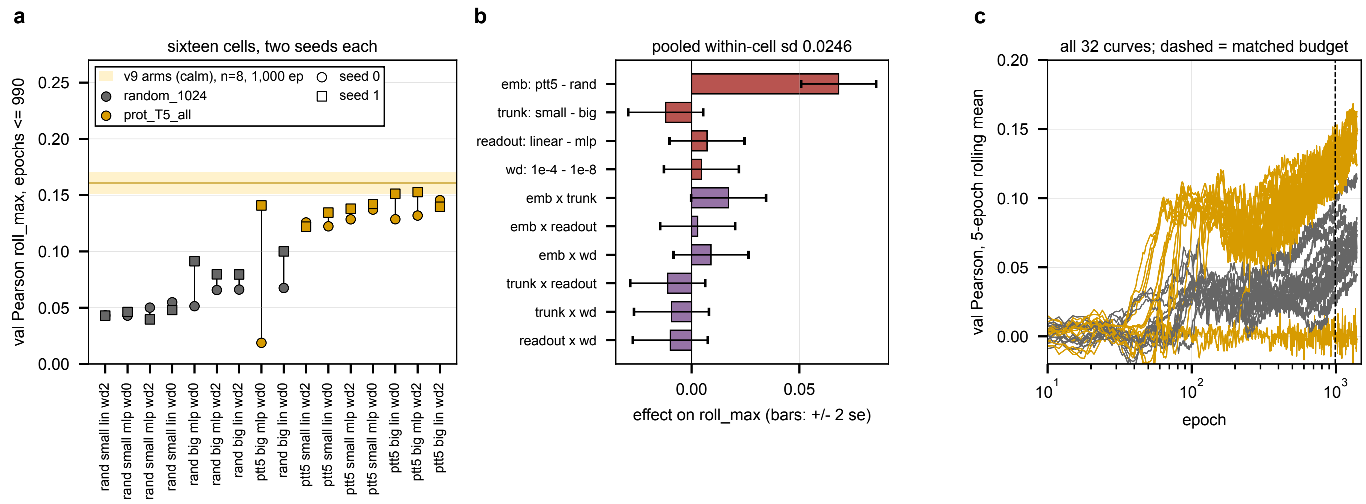

Figure 3: The v10 grid at a matched budget. a, every cell's two seeds at epochs ≤ 990, sorted by cell mean and colored by embedding, against the eight long-budget v9 arms at 1,000 epochs (0.161 ± 0.010, `calm` embeddings, band is one standard deviation across arms). **b**, main effects and two-way interactions with ± 2 standard errors from the pooled within-cell spread. **c**, all 32 validation curves; the dashed line is the matched budget. Generated by `v10_grid_factorial.py`.

`linear_readout` was declared in the config and carried in every run's recorded configuration, but the model never passed it to the head, so both levels trained the two-layer head. The parameter counts logged to W&B are identical across the two levels, 1,447,309 under the deep trunk and 682,988 under the shallow one, which is the tell. The option was wired on 2026-09-10 with a regression test, and linear against two-layer readout is unmeasured until the v12 readout round (section 4). The trunk is a small effect in favor of the deeper one, -0.012 over all runs and -0.018 once the stuck run is dropped, and it is sharper under ProtT5 (the interaction). Of the three factors that were in fact varied, one acts.

One run in 32 never learned. Cell 1 seed 0 (`te8272kk`: ProtT5, L=6, two-layer readout, weight decay 10^{-8}) sits at 0.019 at its best and -0.007 at epoch 1,399 with `nmse` 1.010, while its seed-1 twin is the grid's best single run at 0.169. That pair alone carries most of the pooled variance, which is why both readings are given. Its cause is not measured.

3.2 The mechanism round: per-gene readout +0.03, under the round's resolution ✗

All four array tasks of the round ended in the cgroup out-of-memory handler at host RSS 61 GB against a 60 GB request, after 3.5 to 4.6 days. In each task one of the two packed runs was killed and the other finished its 8,500 epochs, so the matched budget is set by the shortest survivor, epochs $\leq$ 4,079. The two seed-1 runs abandoned on cabbi at epoch $\approx$1,595 when that half moved to the A40 node are superseded, not replicates, and are dropped.

arm	seed 0	seed 1	paired difference vs `R_ref` (s0 / s1)	mean
`R_ref`	0.188	0.169		
`R_basis64`	0.175	0.200	-0.014 / $+0.030$	$+0.008$
`R_pergene`	0.189	0.224	$+0.001$ / $+0.054$	$+0.028$
`R_pergene_basis64`	0.214	0.199	$+0.026$ / $+0.029$	$+0.027$

Table 12: The mechanism round at epochs $\leq$ 4,079, paired within seed against the in-round reference. Seed 0 ran on cabbi Ada cards, seed 1 on one A40 node, so GPU type is a seed-level block. The eight long-budget v9 arms score 0.188 ± 0.017 at 4,000 epochs and both `R_ref` draws fall inside that band.

The pair term alone is a null. Its two paired differences disagree in sign and average +0.008. The per-gene readout arms average +0.027, and the combined arm is positive at both seeds, but the round was sized to resolve about 0.06 at two pairs (table 9), so this is a direction and not a result. What every curve does show is the dynamics: the per-gene arms reach their peak between epochs 1,200 and 2,600 and give part of it back (`R_pergene` ends at 0.137 and 0.170 against peaks of 0.189 and 0.224), while the reference and basis arms are flat or still rising at 8,500. Whether the give-back is overfitting of the extra per-gene parameters is a hypothesis and is not measured.

A tagging defect, found in the readout. The arm script carried a wave-4b `R_ref` branch ahead of the wave-5 one; a shell `case` takes its first match, so every reference run of this round is tagged `stage-wave4b`. The readout selects by arm and config tag, and the shadowing branch is removed.

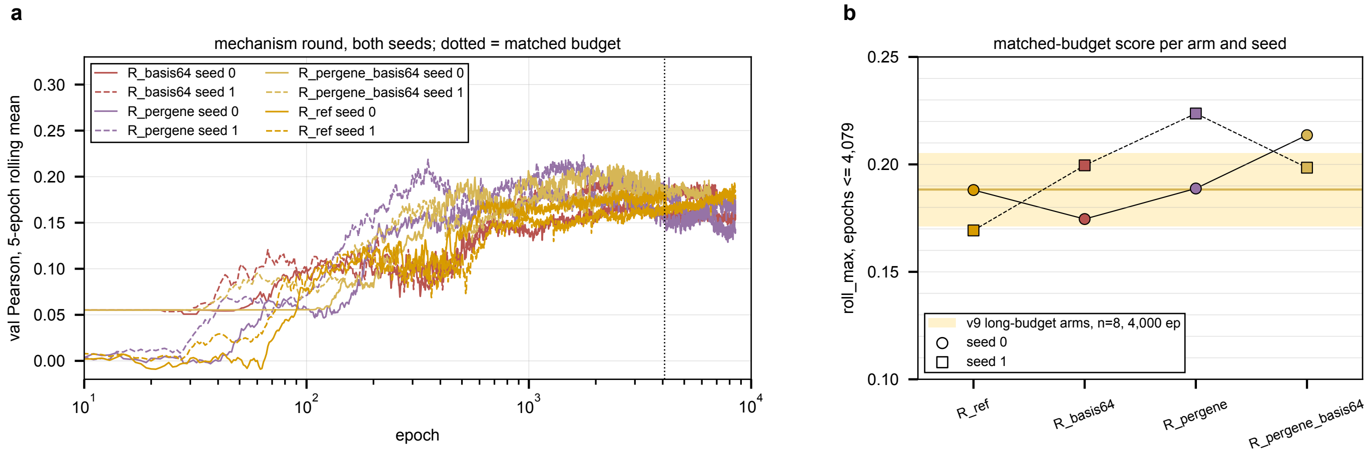

Figure 4: The mechanism round. **a**, validation Pearson of all eight runs (solid seed 0, dashed seed 1); the dotted line is the matched budget. **b**, the matched-budget score per arm and seed against the long-budget arm band at 4,000 epochs. Generated by `mech_round_readout.py`.

3.3 The metric-aligned round at day three: collapse is the rule at batch 32 ✗

Three arms train on $1 - r$ directly, the mean per-feature Pearson over the genes hidden at the current unmasking step (`pearson`), the same plus a masked squared error at unit weight (`pearson_mse`), and the pure loss at batch 64 (`pearson_b64`), on IGB cabbi from 2026-09-06. The two batch-32 arms run three seeds at two runs per card, the batch-64 arm two seeds solo. Five of the six batch-32 runs were on the floor by day two and were stopped on 2026-09-08 (two whole tasks canceled, one dead process killed inside its task so its live sibling kept the card); at the read of 2026-09-09 18:15 CT the three survivors are still training, the batch-64 pair at 9,114 and 9,118 of 9,900 and the anchored seed-1 run at 6,576. The readout script records the epoch each reached, and every number below is partial at that epoch.

arm	seed	epoch	`roll_max` @ epoch	Spearman	last	on the floor from	spread ratio
`pearson`	0	4,298	0.145 @ 215	0.092	0.000	584	0.000
`pearson`	1	4,291	0.133 @ 151	0.085	0.000	2,203	0.000
`pearson`	2	3,166	0.137 @ 242	0.086	0.000	680	0.000
`pearson_mse`	0	4,298	0.137 @ 138	0.086	0.000	360	–
`pearson_mse`	1	6,576*	0.194 @ 3,193	0.178	0.172	never	0.43
`pearson_mse`	2	3,169	0.035 @ 5	0.030	0.000	14	–
`pearson_b64`	0	9,118*	0.194 @ 1,926	0.175	0.148	never	0.89
`pearson_b64`	1	9,114*	0.186 @ 2,419	0.171	0.160	never	0.90

Table 13: The metric-aligned round at day three; * marks a run still training at the read. Spearman is the `roll_max` of the per-feature Spearman, the quantity the ranking round targets. "On the floor from" is the first epoch after which the 5-epoch rolling mean of the validation Pearson stays under 0.02; the seed-1 `pearson` run is near zero from epoch ≈250 and flickers above the line until 2,203. The spread ratio is predicted over true standard deviation (`pred_sd_ratio`) at the last epoch. The eight long-budget v9 arms score 0.188 ± 0.017 at 4,000 epochs, 0.192 ± 0.018 at 6,000 and 0.1965 ± 0.0222 at 9,900.

Five of six batch-32 runs are on the floor, and the loss does not see it. The three pure-Pearson runs peak between 0.13 and 0.145 at epochs 150 to 240 and then fall, their predicted spread dropping from about 0.3 to 10^{-7} over the same epochs (fig. 5b): the network's output goes constant per gene. Their validation loss holds near 0.56 throughout, because the loss drops a constant column as invalid and averages over the columns still varying, so it reads $1 - 0.44$ on a handful of genes while the metric reads 0 on all of them. The failure mode the design named is the one that occurred, with the difference that the loss never reads its connected zero. The squared-error anchor at unit weight did not prevent it: `pearson_mse` collapsed by epoch 14 at one seed and 360 at another, and its surviving seed reads `roll_max` 0.194 at epoch 6,576 with a last value of 0.172, inside the long-budget arm band and still training.

The batch-64 runs survive and sit on the band, not above it. At 9,114 and 9,118 epochs, with spread ratios of 0.89 and 0.90, they read `roll_max` 0.194 and 0.186 against the incumbent's 0.1965 ± 0.0222 at 9,900, peaked near epochs 1,900 to 2,400 and sit at 0.148 and 0.160 now, 0.03 to 0.05 below their peaks with under 800 epochs to run. Batch size and solo placement are confounded in that arm, so why it survives is not measured; the in-batch correlation being taken over twice as many strains is a hypothesis. Nothing in the round beats the quantile head at any budget read so far. The question it was launched to answer, whether the metric's late rise is generalization gap or objective disagreement, is not answered by two survivors that peaked by epoch 2,400: on the Pearson objective the late rise did not happen, and whether that is the objective or the batch size is the confound above.

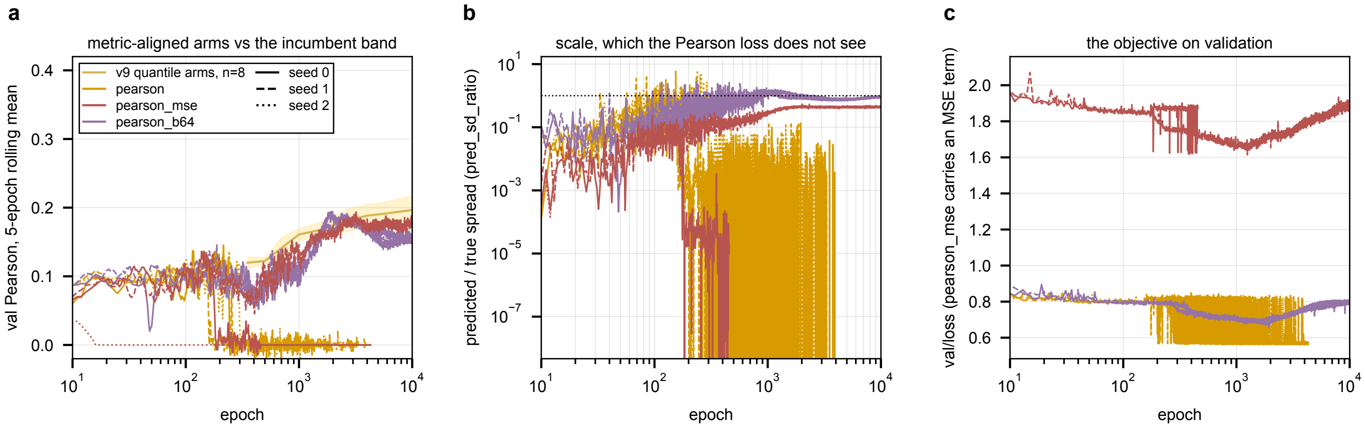

Figure 5: The metric-aligned round at day three. a, validation Pearson of all eight runs against the long-budget arm band. **b**, predicted over true spread, which the Pearson loss cannot see; a run whose outputs go constant falls off the bottom. **c**, the objective on validation; `pearson_mse` carries the squared-error term and sits on its own scale. Generated by `pearson_round_readout.py`.

3.4 The ranking round at day one: ListMLE learns, slowly, and does not collapse ✗

Three arms replaced the dead half of the metric-aligned round on 2026-09-08 20:05 CT: `listmle` at batch 32, two seeds packed on cabbi with the anchored `listmle_mse` beside them, and `listmle` at batch 64 solo on two A40s, all at a 6,000-epoch budget chosen because every packed five-day task so far died of host memory between days 3.5 and 4.6 and the Pearson peaks had come by 3,200. Read at 2026-09-09 18:15 CT with every surviving run still training (`pearson_round_readout.py--roundlistmle`).

arm	seed	epoch	`roll_max` @ epoch	Spearman	last	on the floor from	spread ratio
`listmle`	0	1,959*	0.168 @ 1,920	0.173	0.167	never	0.80
`listmle`	1	1,792*	0.150 @ 1,488	0.154	0.140	never	0.75
`listmle_b64`	0	2,438*	0.144 @ 2,433	0.134	0.137	never	0.67
`listmle_b64`	1	2,437*	0.150 @ 2,057	0.138	0.144	never	0.67
`listmle_mse`	0	637	0.025 @ 1	0.024	0.000	3	0.000
`listmle_mse`	1	480	0.024 @ 246	0.024	−0.033	247	0.000

Table 14: The ranking round at day one; * marks a run still training. Columns as in table 13. The eight long-budget v9 arms score 0.1745 ± 0.0151 at 2,000 epochs and 0.1821 ± 0.0175 at 2,800. The two anchored runs sat at the mean predictor from epoch 160 and were killed by process at 03:40 CT on 2026-09-09 so their pure siblings could take the whole card.

The anchor kills it, the pure loss does not. With the squared error at unit weight both seeds went to the per-gene mean by epoch 160 and stayed there; the MSE gradient is about 50× the ranking gradient at initialization and reaches its own optimum, the constant, which the ranking term's $\log(n!)/n$ floor does not pull it off. That is the mirror image of the Pearson round, where the anchor was what kept the one surviving batch-32 run alive. The pure ranking loss has not collapsed at any of its four runs, and its spread ratio climbs, from under 0.05 before epoch 200 to 0.80 at 1,959, as a shift-invariant objective that rewards larger score gaps should make it; the leaderboard's `nmse` of 74.7 for these runs (table 8) is that free location, not a failure to order.

Behind the band at matched epochs, by 0.01 to 0.04. At epochs $\leq 2{,}000$ the batch-32 pair reads 0.168 and 0.150 against the long-budget arms' 0.1745 ± 0.0151, one seed inside the band and one below it; the batch-64 pair at 2,400 reads 0.144 and 0.150 against 0.1821 ± 0.0175 at 2,800, both below. Every curve is still rising and none has peaked, so these are epochs reached and not results. On the quantity ListMLE targets, the per-feature Spearman, the batch-32 seed-0 run reads 0.173 at 1,920 epochs where the surviving Pearson-objective runs read 0.171 to 0.178 at 6,500 to 9,100; whether that is a real ordering advantage or a matched-epoch artifact is answerable when the round ends. Hypothesis (untested): the ranking gradient is small in absolute terms, norm 0.3 at initialization, and the learning rate pinned at 3×10^{-4} for the pinball loss is low for it, which would show as the slow start of fig. 6a.

3.5 Packed five-day runs die of host memory ✗

Four of four packed tasks that ran past three and a half days ended `OUT_OF_MEMORY` (`2369697` 0–1, `2371531` 2–3), at `MaxRSS` 61.4 and 61.1 GB. The Pearson-round packed tasks read 21.3 GB at 1.9 days and the solo batch-64 tasks 17.2 GB.

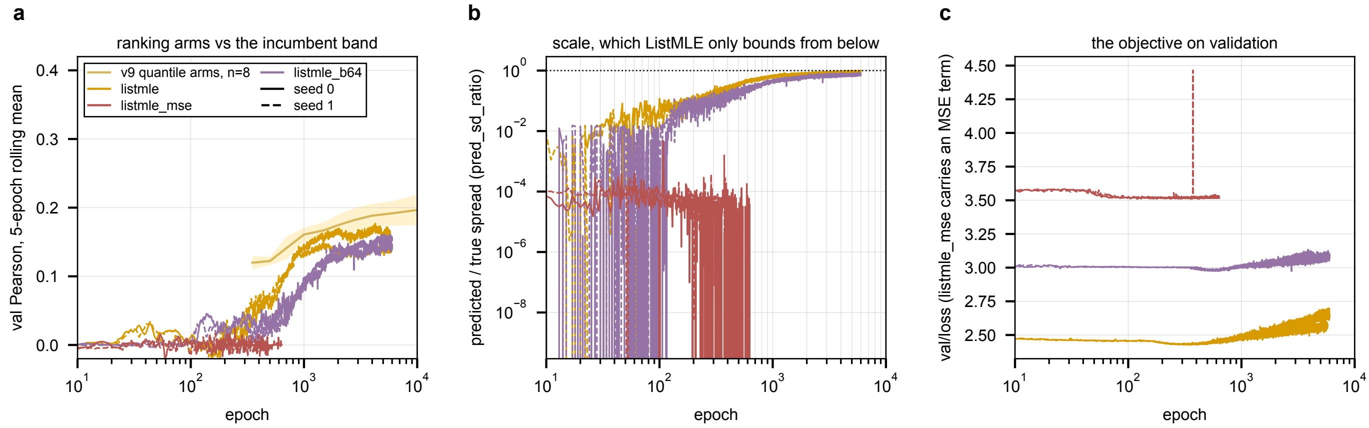

Figure 6: The ranking round at day one. **a**, validation Pearson of all six runs against the long-budget arm band. **b**, predicted over true spread; the anchored runs fall off the bottom by epoch 250 while the pure runs climb from 0.05 toward 1. **c**, the objective on validation, which bottoms at epochs 290 to 630 and rises after, as every other objective in fig. 2 does. Generated by `pearson_round_readout.py--roundlistmle`.

Hypothesis (untested): resident memory grows roughly linearly through a run, in which case the packed Pearson tasks reach the 60 GB line near day five, at the edge of their wall. What grows is not identified; the `file_system` sharing strategy introduced in `f1bfa95b` to stop the file-descriptor exhaustion is the obvious suspect and is unverified. The memory of a running job cannot be raised, and the two rounds that finished lost one run per task to it rather than the whole task.

3.6 A third deletion panel, and whether it measures the same thing ✗

Every round above trains and scores on the Kemmeren and Sameith microarrays, whose perturbation axis has 1,484 points (section 1.1). Nadal-Ribelles et al. 2025 (doi:10.1038/s41467-025-57600-4) profiled about 3,100 single deletions of the same collection by single-cell RNA-seq, pseudobulked per genotype against a wild type in the same run, in base medium and under 0.4 M NaCl. The base-medium records are 3,039 deletion ORFs, of which 914 are in Kemmeren and 44 in Sameith; the union of the three panels is 3,609 deletions, 2.4 times today's axis. Before that panel can extend the axis it has to agree with the other two on the deletions they share, and fig. 7 measures that with the same test-retest construction as the ceiling, on the base-medium records only, with Kemmeren against Sameith as the reference for what two studies of one platform look like. SRC: `experiments/028-knockout-expression/results/cross_study_ko_expression.json`

The stored values are on another scale, and one cell in five is a sentinel. The microarray $\log_2$ ratios have standard deviation 0.23 and 0.18; the stored Nadal-Ribelles values have 10.1 as stored and 0.89 after the correction below (fig. 7a, b). They are scanpy `logfoldchanges` from a Wilcoxon rank-sum call, computed as $\log_2$ of the ratio of two expm1-transformed means with a 10^{-9} pseudocount, which returns ± 23 to ± 33 whenever one group's mean is zero: 20.2 % of the stored cells carry such a value and 22.6 % of genes carry it in more than half of their records. The Wilcoxon test is the paper's DE test; the fold change beside it is a separate quantity, and the paper used it only above an independent filter at mean normalized counts of 1.27 and a p threshold, whereas the loader stored it for every gene with a nonzero count. Those cells are treated as absent here. Even after that, the per-strain spread of a Nadal-Ribelles profile is bimodal, with a mode at sd 0.86 against the microarrays' 0.19 and 0.15 (fig. 7c), so the remaining values are still not on the microarray scale.

On the shared deletions the two panels almost do not agree. Over 914 shared deletions the per-strain profile correlation with Kemmeren has median 0.003, against 0.744 for Kemmeren with Sameith over 82; the per-reporter test-retest median is 0.030 against 0.620; the OLS slope of Kemmeren on Nadal-Ribelles over 4.06 million shared cells is 0.006 against 0.82 on Sameith (fig. 7d–f). On the reporters where Kemmeren's effect exceeds one $\log_2$ unit the agreement is weak but above chance, median 0.10 with signs agreeing 60 % of the time (chance 50 %, the microarray pair 93 %). The gene mapping is right: the cys3 deletion carries MET14, MET3 and MET6 up by 3 to 3.7 and CYS3 itself down, the induction that deletion is known for. The paper's only comparison to Kemmeren is a count of genes at $p < 0.05$ over 837 shared genotypes; it reports no profile correlation.

What the agreement tracks. It rises with the strain's effect size in Kemmeren (Spearman 0.37 over 914 strains) and with the reporter's variance across strains (0.29 over 5,405 reporters), and it does not track the number of cells behind the pseudobulk at all (Spearman 0.02; median 92 cells per genotype) (fig. 7g–i). A pseudobulk that was merely undersampled would agree better where more cells were pooled; this one does not.

Recomputing the fold change from the cells does not recover agreement. The single-cell object (`seus_split.RData`, Zenodo 14062629, now in the raw mirror) holds 326,438 control cells over 3,506 genotype labels with raw UMI counts, so the statistic can be replaced. Three were computed per genotype against the 500 pooled wild-type cells: a bulk-like pseudobulk ratio, summed UMI to counts per million, $\log_2((\text{cpm}_g + 1)/(\text{cpm}_{wt} + 1))$; Seurat's `avg_log2FC`; and scanpy's own formula, which reproduces the stored sentinel rate (22.6 % against 20.2 %, cell-by-cell sentinel status agreeing 98 %) and confirms what was stored. Against Kemmeren over 1,002 shared deletions the per-strain median is 0.013, 0.008 and 0.011 for the three, the per-reporter median 0.019, 0.020 and 0.012, and the slope 0.002, 0.010 and 0.000 (fig. 8a–e); none tracks cell count. The statistic was not the cause. SRC: `experiments/028-knockout-expression/results/cross_study_recomputed.json`

The conditions differ in more than the medium. Kemmeren and Sameith grew each strain alone in SC with 2 % glucose to early mid-log at OD_{600} 0.6, two independent cultures, dye-swapped against a wild-type pool grown the same day, BY4742 MATα. Nadal-Ribelles recovered the URA3-marked collection on URA$^-$ at 25 °C for 48 h, grew each strain 6 h in YPD to OD_{660} 0.6 to 0.8, then pooled all 3,500 strains into one flask and fixed them in methanol; the reference is 500 wild-type cells from seven barcoded wild-type strains in the same run. Kemmeren's own SC-against-YPD comparison of wild type found 128 genes changed at 1.7-fold, so the baseline differs, and a deletion's response can differ with it. Hypothesis, unmeasured here: the medium and the pooled growth lower cross-study agreement below the 0.62 same-platform reference. Whether they take it to zero is what the cell-level measurements below address, and those do not depend on Kemmeren.

Read-through from the barcode cassette is not what puts counts on the deleted gene. The redesigned cassette shortens the heterologous terminator from 262 to 43 nt so that the pTEF1-driven URA3-barcode transcript ends at the deleted gene's own terminator, and 3′-end capture reads that transcript at the deleted locus. If that were the source of the deleted gene's counts in its own cells, those counts would sit at one pTEF1 level whatever wild type expresses there. Measured over 2,724 genotypes with at least 20 cells, the fraction of a genotype's cells detecting its deleted gene tracks the wild-type fraction all the way down (Spearman 0.64) at about 0.6× in every bin, and for the 1,586 deleted genes wild type itself detects in under 5 % of cells the genotype's own cells detect it in a median 2.9 %. The counts at the deleted locus scale with native expression; the cassette does not supply them. SRC: `experiments/028-knockout-expression/results/nadal_readthrough_and_signature_tests.json`

Where a response is detectable, about half the labeled cells carry it; for most genotypes none is detectable. Two single-cell tests that never use the deleted gene. For the 240 shared deletions with at least 20 Kemmeren reporters beyond one $\log_2$ unit, every cell of the genotype and every wild-type cell was scored on Kemmeren's signature (mean over up reporters minus mean over down reporters). The score separates the genotype's cells from wild type at a median AUROC of 0.53 (IQR 0.50 to 0.61), 8 of 240 above 0.8; a median 7.5 % of a genotype's cells sit above the wild-type 95th percentile, against 5 % by chance. The 8 that separate are the strong regulators, *rfx1*, *rim101*, *skn7*, *sam1*, *ada2*, *snf5*, *swi3*, YGR122W, and in those 33 to 68 % of the labeled cells sit above the wild-type 95th percentile, the rest inside the wild-type range. The second test asks for any reproducible signal: each genotype's cells were split in half, a 50-gene signature derived from one half against half the wild-type cells, and the other half scored against the held-out wild-type half. Over 207 genotypes with at least 40 cells the median AUROC is 0.52 (IQR 0.49 to 0.57) against a null of 0.49 (wild-type cells playing a genotype), with 18 above 0.65 and 6 above 0.8; the two tests agree on which genotypes separate (Spearman 0.24). SRC: `experiments/028-knockout-expression/scripts/nadal_signature_mixture.R,nadal_split_half_signal.R`

The excess-zero purity estimate (median 0.36 to 0.40 from the deleted gene's own counts; fig. 8f) and the signature fractions above say the same thing for the genotypes with a strong response: roughly half of the cells under a label are that genotype. For the majority of genotypes the labeled cells carry no reproducible signal at this depth (a median of 92 cells at 215 to 7,713 UMI each), and whether that is the same impurity, a response too small for the depth, or a response absent in YPD and pooled growth is not separable here. Batch is not the cause; 3,307 of the 3,506 genotypes span three or more of the 14 batches and the wild-type cells span all of them. What produces the impurity in the strong-response genotypes, the clone-and-genotype barcode consensus, ambient barcode reads in the microwell format, or strain identity in the library, is not measured here.

The deleted gene cannot be read off its own profile. A deletion removes its own transcript, so in a clean knockout profile the deleted gene sits at the bottom. In Kemmeren it is the single most reduced gene in 59.5 % of 1,479 strains and in the bottom 1 % in 93.4 %, with a median own value of −2.48. In Nadal-Ribelles it is at rank 1 in 0.1 to 0.8 % of genotypes and in the bottom 1 % in 2.0 to 8.2 % under the three statistics, with a median rank at the 35th to 47th percentile of the profile (chance is the 50th) and a median own value of −0.02 to −0.41 (fig. 9). The rate does not rise with cells per genotype (2.0 % under 30 cells, 0 % above 300). The deleted gene is not identifiable from a Nadal-Ribelles profile, and read-through from the cassette is not the reason (above). SRC: `experiments/028-knockout-expression/results/nadal_identify_deletion.json`

As released, the panel cannot be pooled with the microarrays as a knockout-expression target: on the shared deletions the pseudobulk agrees with Kemmeren at zero under every statistic, the strong responses that do cross survive at 0.06 to 0.10, and more cells do not help. Where the labeled cells are half another genotype, a per-cell filter on the deleted gene's own counts (the cells at zero) or a re-assignment from the raw barcode reads would be needed, and the released object carries only the consensus label. Its per-genotype dispersion, a single-cell heterogeneity readout the loader already carries, is a different target and is not affected in the same way.

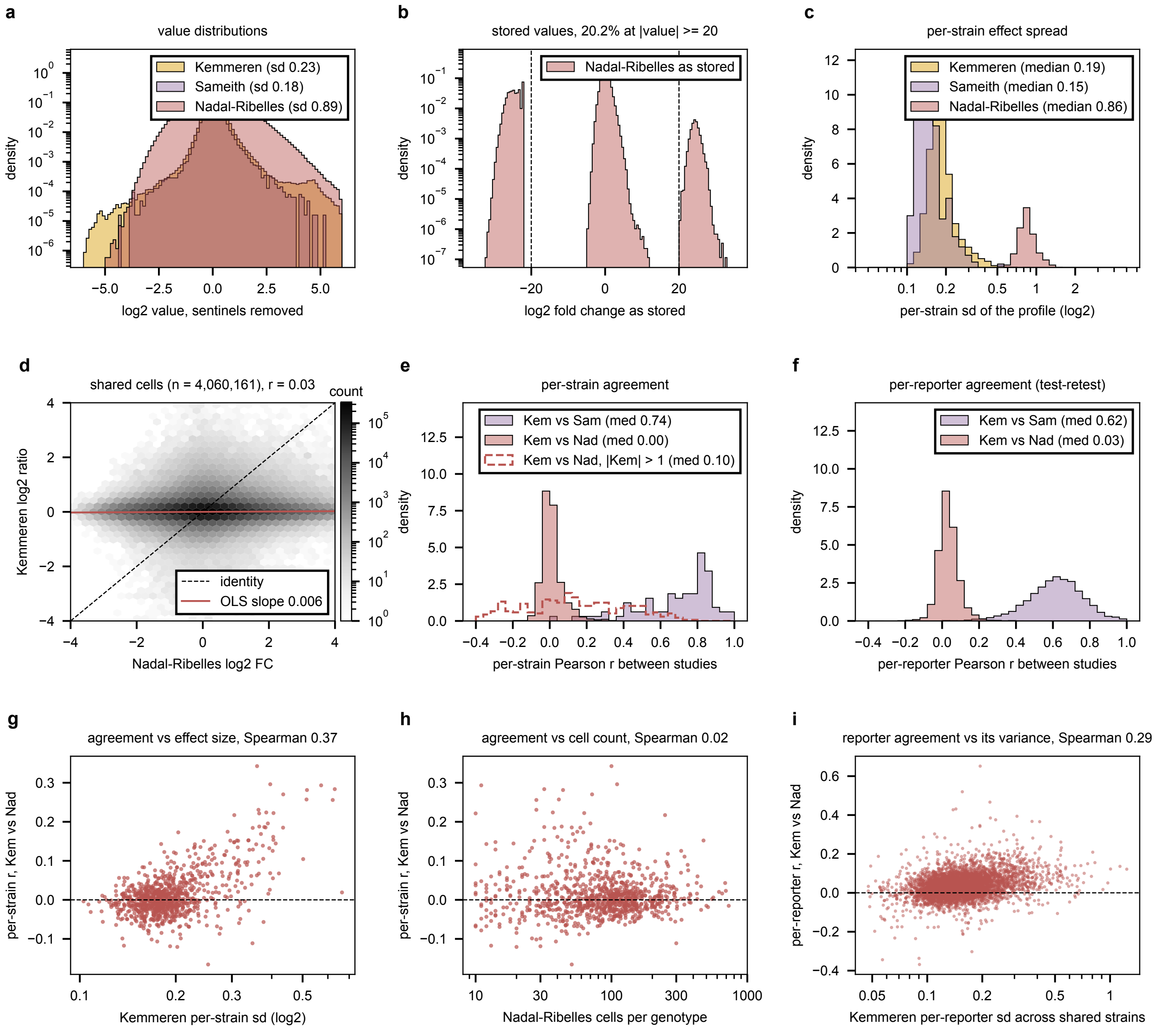

Figure 7: Three single-deletion expression panels compared, base medium only. **a**, $\log_2$ values of each panel with the Nadal-Ribelles sentinels removed; the two platforms sit on different scales. **b**, Nadal-Ribelles values as stored, with the ± 20 lines: 20.2 % of cells are the zero-mean sentinel. **c**, per deletion, the standard deviation of its profile across reporters. **d**, Kemmeren against Nadal-Ribelles on every shared (strain, gene) cell, log-count density, with the identity and the fitted slope. **e**, per shared deletion, the Pearson across reporters between its two profiles: Kemmeren vs Sameith ($n = 82$), Kemmeren vs Nadal-Ribelles ($n = 914$), and the latter over only the reporters with $|\log_2| > 1$ in Kemmeren (dashed, $n = 368$ strains). **f**, per reporter, the Pearson across shared strains between its two response vectors, the test-retest reliability behind the ceiling. **g**, the per-strain agreement of e against the strain's Kemmeren spread; **h**, against the number of cells behind its pseudobulk; **i**, the per-reporter agreement of f against the reporter's Kemmeren spread across the shared strains. Generated by `cross_study_ko_expression.py`.

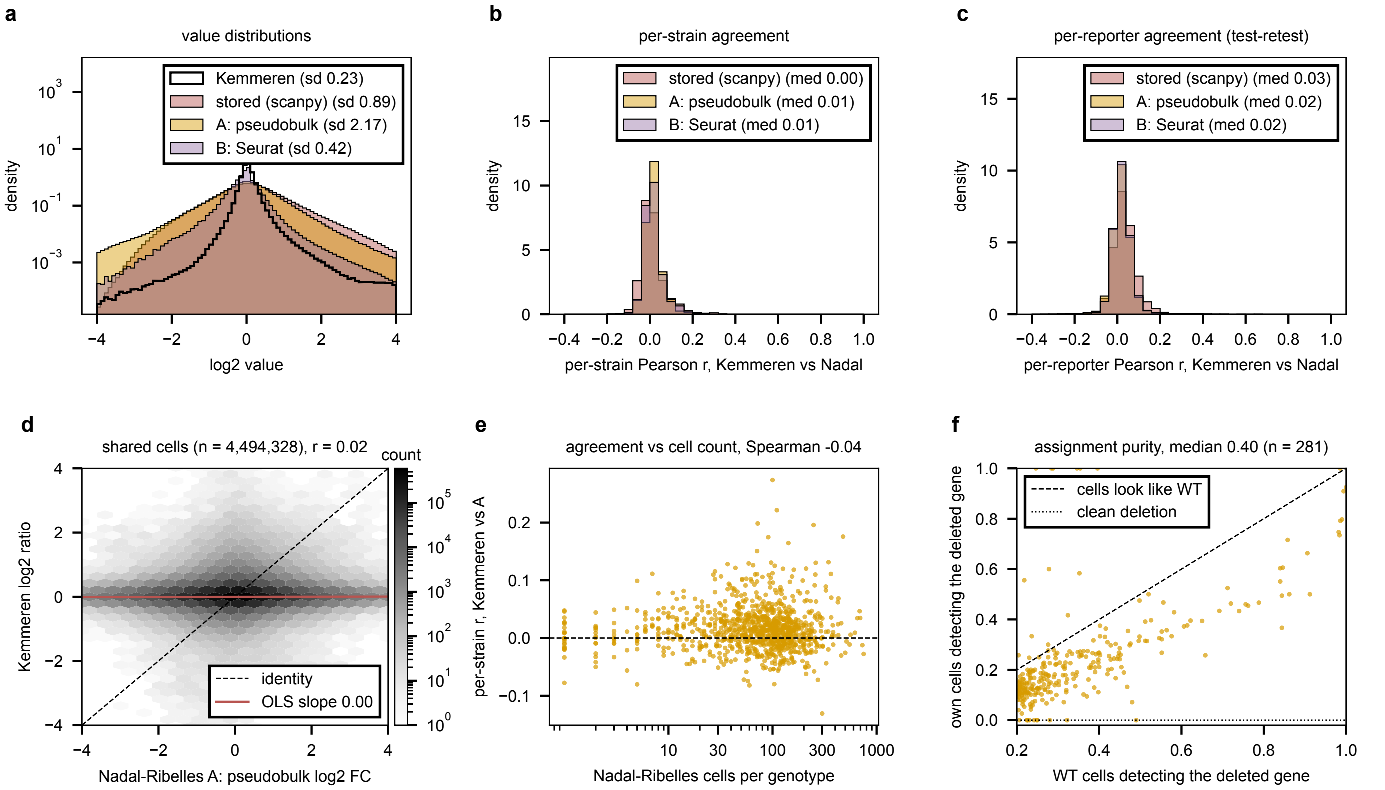

Figure 8: Recomputing the Nadal-Ribelles fold change from the single-cell object, and why it does not help. **a**, $\log_2$ values of Kemmeren (outline) and of three Nadal-Ribelles versions: as stored, the bulk-like pseudobulk ratio (A) and Seurat's `avg_log2FC` (B). **b**, per shared deletion, the Pearson across reporters with Kemmeren for each version; **c**, the same per reporter across shared strains. **d**, Kemmeren against A on every shared cell. **e**, per-strain agreement of A against the cells behind the genotype. **f**, assignment purity: for each genotype whose deleted gene wild type detects in at least a fifth of cells, the fraction of the genotype's own cells detecting that gene against the wild-type fraction; the dashed identity is what cells that are not the genotype look like, the dotted zero a clean deletion. Generated by `cross_study_recomputed.py` from `nadal_pseudobulk_recompute.R` and `nadal_assignment_purity.R`.

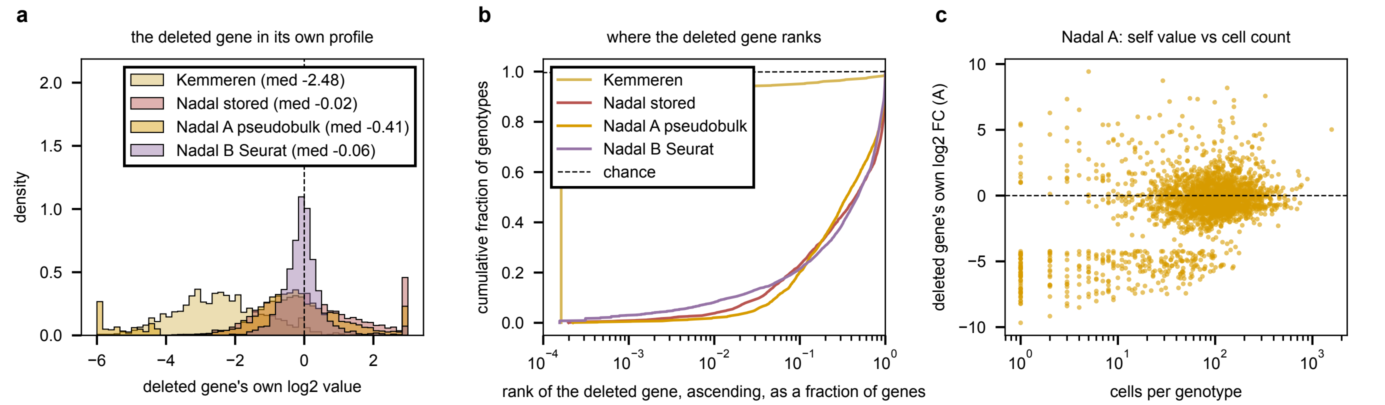

Figure 9: The deleted gene in its own profile. **a**, the deleted gene's own $\log_2$ value per genotype for Kemmeren and three Nadal-Ribelles statistics. **b**, cumulative distribution of the deleted gene's ascending rank as a fraction of genes in the profile, log axis; the dashed diagonal is chance. **c**, for the pseudobulk statistic, the deleted gene's own value against the cells behind the genotype; the band near -5 to -8 at low counts is the pseudocount floor of a near-empty pseudobulk. Generated by `nadal_identify_deletion.py`.

4 What the evidence leaves open, and what fits the hardware ✗

Each item is a decision the measurements above do not make, stated with the measurement that would. The hardware they have to fit, as of 2026-09-09 18:10 CT: on IGB, all eight cabbi RTX 6000 cards are allocated (three to another user, five to this campaign's surviving runs), the two batch-64 Pearson tasks free two cards when they reach 9,900 in about six hours, the packed anchored-Pearson task frees one about a day later, and the ranking round holds two cabbi cards and two A40s for one to two more days; four A40s on the `gpu` partition are idle; the mmli A100s are not to be touched. On Delta the `bbub` allocation holds 3,738 GPU-hours under a 48-hour wall, which at the v10 grid's measured rate of 990 to 1,400 epochs per task fits every 1,000-epoch design below and none of the 9,900-epoch ones.

Which number is the target. Stopping on the validation loss says every run is done by epoch 300 to 700 and the long-budget campaign is over; stopping on Pearson says the curves have not turned at 19,000 (section 2.5). The manuscript reports Pearson and the objective is the pinball loss, and the two metric-aligned rounds show the naive resolution is unstable at batch 32 under Pearson, no better than the incumbent at batch 64, and slow under ListMLE (section 3.3, section 3.4). The decision is about what the paper claims, not about what to run.

The incumbent's embedding against ProtT5: the cheapest contrast on the table, and it fits Delta. The grid measured content against random at width 1,024 and did not include `calm` (section 3.1). A two-arm replicate set at the incumbent configuration, `calm` against `prot_T5_all`, three seeds each at 1,000 epochs, resolves a gap of about 0.03 at the within-cell replicate sd of 0.012 to 0.025, and is six runs of under 48 hours each: one Delta array on `bbub`, about 300 GPU-hours. It also gives the first true replicate spread of the incumbent configuration, which the eight long-budget arms only bound, and that spread is what every later design is sized by.

The per-gene readout at a resolving replicate count, also on Delta. +0.027 at two pairs against a resolution of 0.06 (section 3.2). Detecting 0.03 at the 4,000-epoch spread of 0.0171 needs about six pairs; at 1,000 epochs and 0.0099 it needs three, which is six runs, `R_ref` against `R_pergene`, and a second 48-hour Delta array. Whether the arm's early peak and give-back survive a shorter budget is what the same runs show.

The readout round, submitted 2026-09-10 on IGB. The published readouts reduce to a short list once the shared trunk is fixed: GEARS's gene-specific row on the gene's own token, that row plus GEARS's pooled cross-gene state, State's $\mathbf{W}_{\text{recon}}$ and the benchmark's ridge decoder (a gene-specific row on the strain vector), scGPT's masked-value form (a factored bilinear, gene sensitivity times strain amplitude), State's embedding-model decoder (a shared MLP over the gene embedding and the cell vector), and the shared linear map. Each maps to one existing option of the per-gene head except the strain-vector row, added as `context_readout` and zero-gated like the others. Eight arms by four seeds, four runs per 48 GB card (a batch-32 run peaks near 9 GB on the W&B system metrics), seed-major so every contrast against the shared-MLP reference is paired within seed with the card as a block, on the full-locus input of section 4's embedding round under the pinball objective, at that round's 1,400 epochs so the reference arm replicates its widest arm: jobs `2392379` (cabbi) and `2392381` (`gpu`), source `271b2c71`. The wiring of `linear_readout` was found broken while building it (section 3.1), so the linear arm is its first measurement. Four pairs per contrast at a paired sd near 0.03 resolve about 0.035; the full-locus input is itself unmeasured against `calm` until the embedding round lands, and every contrast here is within it.

The batch-32 solo Pearson run that separates batch size from placement. The batch-64 pair are the round's only informative survivors and batch size is confounded with solo placement in that arm (section 3.3). One batch-32 solo seed of the pure objective is the run that separates the two; it needs the 9,900-epoch budget, so it takes one cabbi card for 3.5 days when the batch-64 tasks free theirs.

The ranking round, when it ends. The four surviving runs reach 6,000 epochs in one to two days. The readout then is the matched-budget Spearman and Pearson against the long-budget arms at 6,000 (0.192 ± 0.018) and against the Pearson-objective survivors, and the one follow-up the current curves motivate is a learning-rate arm, since the slow start is the only thing separating ListMLE from the band so far and the rate was inherited from the pinball loss. That is a hypothesis until the round reads out.

The perturbation never touches the encoder, and the CLS state is wild-type. This is the structural fact of section 1.2: the encoder runs once on the unperturbed graph, h_{CLS} is byte-identical across strains, and the deletion is applied to finished gene states through a one-key cross-attention. Every mechanism arm so far has been a different way of applying the perturbation after the encoder, and none puts it inside. Three designs would, none of them trained here, listed by how much they change:

- **Perturb the CLS as well as the genes.** Run the existing perturbation operator on h_{CLS} too, cross-attending it to the deleted set, so the whole-cell summary becomes strain-specific at the cost of one more query. Cheapest, restores a strain-dependent global state, and leaves the pair-term degeneracy exactly where it is.

- **Edit the token before the encoder.** Remove or flag the deleted gene's token in the encoder input (Geneformer's in-silico deletion, scGPT's perturbation flag), so the graph-masked self-attention carries the deletion outward to every gene and to CLS along the interaction graphs, and the one-key degeneracy never arises. This is the design that uses the graphs for the perturbation rather than only for the wild-type state. Its cost is one encoder pass per strain instead of one per batch; hypothesis (untested), at 39 batches per epoch and 17 to 31 s/epoch that is a wall-clock multiplier near the batch size unless a shallow strain-specific encoder is stacked on frozen wild-type layer outputs.
- **Perturbation tokens, typed by what is known.** The token-edit design works because a deletion names its node. An environmental perturbation, an antifungal added to the medium, names no gene, so it cannot be an edit; the generic form is a set of typed perturbation tokens entering the encoder beside the gene tokens, a gene-edit token that attends to its gene and an environment token that attends to CLS or to every gene, with the encoder deciding which genes an environment token reaches. That covers deletions and environments in one mechanism and makes CLS strain-specific by construction, and it is the same encoder-per-strain cost as the token edit.

None of these is launchable this week: the first is a small change to the model file, the other two restructure the forward pass, and all three are hypotheses about mechanism until run at a matched budget against the incumbent.

Packing. Four of four packed five-day tasks died of host memory (section 3.5). Until the growth is identified, a five-day arm is either solo on its card or budgeted at 48 GB per run rather than 30, and either choice halves the slots the launch arithmetic assumed. Delta's 48-hour wall sidesteps it for the short designs.

The per-gene mean on squared error. No live run beats `nmse` $= 1$ at its Pearson peak (section 2.1). The one-parameter rescale argument of section 1.2's source, multiplying every prediction by r/s estimated on train, changes no correlation and takes `nmse` to $1 - r^2$; it has not been applied to any run of the campaign, costs nothing, and belongs at the point a figure is drawn.